ASSIGNMENT 3

Demon Slayer Corps — The String Hunt

String matching in practice: Naïve · Rabin-Karp · KMP · Finite Automata

> *In memory of Akaza, Upper Moon Three. It is only fitting that Assignment 3 is his.*

Due: Sunday 8 November 2026, 11:59 PM

Orders from headquarters

Congratulations, you survived Final Selection and joined the Demon Slayer Corps. A demon is hiding among the villages, someone in the Corps is cheating, and the Hashira want to know which technique is fastest. Your sword this time is the string matching: naïve, Rabin-Karp, KMP and finite automata.

Clear three missions and hand in a short **mission report**. All data files and the Python starter code are in `DemonHunt_data.zip` on the GCR. The starter code is in Python; if you prefer C++ or Java, port it yourself.

Start here. The starter code already reads the files, times your runs and tests your work. Your job is to turn the lecture pseudocode into working code: fill the TODOs in `starter/matchers.py`, run `python test_matchers.py` until every line says PASS (your own Final Selection), then complete `mission1.py` to `mission3.py`.

Corps rules

- Language: C++, Java or Python.
- No built-in substring search or regex inside any matcher. Using one in a matcher scores zero for that matcher.
- Report each match as its shift s, exactly as in the lectures ($0 \le s \le n - m$). In Python this is simply the index where the match starts, e.g. `ababa` with `aba` gives 0 and 2.
- Code is checked for similarity with other submissions and online sources. Copied code scores zero for the whole assignment, as per course policy.
- A 10-minute viva may be held. Students who cannot explain their own code or trace their own π and δ may have their marks scaled down.

The three missions

Mission 1 — The Crow's Alarm (KMP) · 30 marks

Every slayer's Kasugai crow reports to headquarters, and all their messages are in `crow_log.txt` (about 5 MB). When a crow spots a demon it screams `KAKAKA`. Headquarters needs to know how many alarms were raised.

- Report how many times `KAKAKA` appears and its first 10 shifts. Overlapping caws count: `KAKAKAKA` holds 2.
- **The catch:** the message scroll is too long to open at once. Read it only in **4 KB chunks**. A caw split across two chunks must still be heard, so carry the KMP state q from one chunk into the next.
- Print the π array of `KAKAKA` as part of your output.

Mission 2 — Tamayo's Blood Lab (finite automaton) · 30 marks

Tamayo has isolated a marker that only demon blood carries: `GATTACAGATTC`. The folder `blood_samples/` holds blood DNA from 10 villagers (about 1 million bases each). One of them is a demon in disguise.

- Build the δ table for the marker over Σ = {A, C, G, T} and print it.

- Run the automaton on every sample and unmask the demon.
- **The catch:** DNA has two strands, and this demon is clever. The marker may sit on the reverse strand only. Also search for the **reverse complement** (swap A↔T and C↔G, then reverse the string). Report matches per strand for every sample.

Mission 3 — The Copycat Slayer (Rabin-Karp) · 30 marks

The Hashira asked five trainees, Tanjiro, Zenitsu, Inosuke, Kanao and Genya, to write about what they learned. Someone found `hashira_manual.txt` and copied from it. Their reports are in `trainee_reports/`. Find out who copied, and how much.

- Clean every text: lowercase, remove punctuation, single spaces.
- Hash every 40-character window of the manual into a hash map (hash → list of shifts), using d = 256 and q = 1,000,000,007. Roll the hash across each report; on a hit, verify character by character.
- For each trainee, print a **similarity score** = copied characters ÷ report length, as a percentage, and rank all five.
- **The catch:** one trainee changed words while copying. In two lines, explain why your detector gives them a lower score than they deserve.

Bonus mission — Breathing Styles Tournament · +10 marks

Optional, for those who want Hashira rank. Race all four techniques (naïve, Rabin-Karp, KMP, automaton) in one arena with `mission4.py`.

- **Arena (Infinity Castle):** T = `a` repeated 100,000 times, P = `a` repeated 99 times followed by `b`.
- Show the comparisons and time of all four in one bar chart, and explain the winner and the loser in 3 lines.

What to submit

Upload one zip named `RollNo_DemonHunt.zip` containing:

- `src/`: your completed `starter` folder.
- `mission_report.pdf`: 1–2 pages with a screenshot of each mission's output and your two-line answer from Mission 3. Add the bonus chart if you did it.

Marks

Part	Marks
Final Selection: `test_matchers.py` all PASS	10
Mission 1: The Crow's Alarm	30
Mission 2: Tamayo's Blood Lab	30
Mission 3: The Copycat Slayer	30
Total	**100**
Bonus: Breathing Styles Tournament	+10